



Data Driven Education

Discussion Forum Subteam

Presentation 3 - 04/16/2025

Meet the Team



Bo Zimpel
Education Researcher



Zilu Zhu
Project Manager/Backend Developer



Diya Jain
Machine Learning Engineer



Saloni Agshiker
Data Scientist



Ethan Lu
Data Engineer

Overview

Introduction

- Online Discussion platform is crucial in modern education.
- **Ed Discussion:** a widely used online discussion forum
 - enables students and instructors to interact
 - discuss course materials asynchronously
- Questions answered:
 - What to do to better utilize Online Discussion platforms?
 - Generate synthetic data
 - Topic Modeling
 - Improve the whole education experience
 - Understand topic-level confusions
 - Enhance course design



Dataset

- Two datasets are used for topic modeling

Ed Dataset

Collected from real courses and students

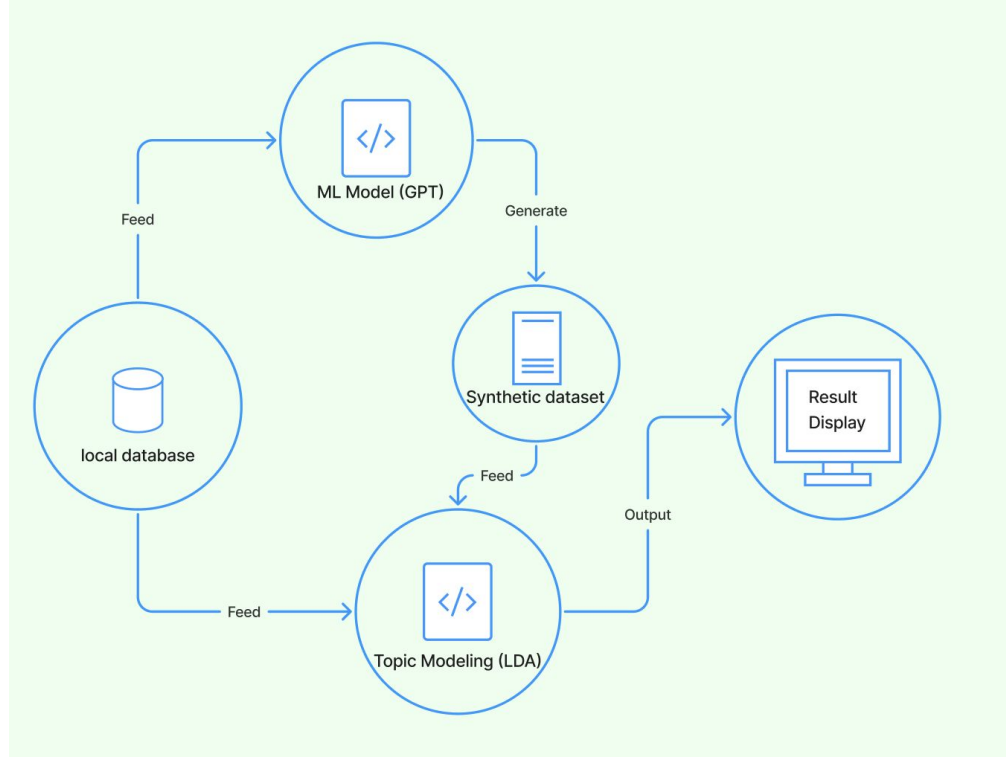
Filtered and cleaned for analysis

Synthetic Dataset

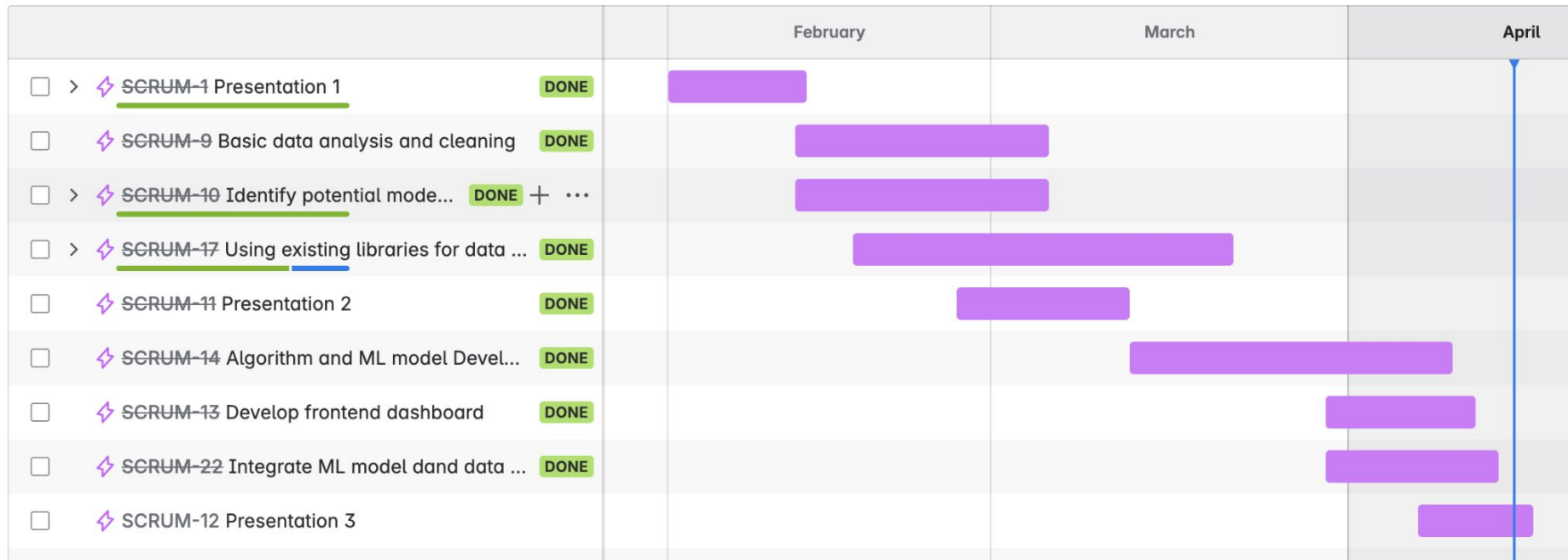
Generated using LLM, same function, more ethical

Cleaned for analysis

Architecture Diagram / Workflow



Project Timeline



Literature Review

Education Research Findings

Initial Goals:

- Identify replicable NLP pipelines for topic modeling and sentiment classification in online discussion forums
- Explore synthetic data generation to supplement real data for model training
- Design dashboard tools that help instructors take action based insights

What Literature Says:

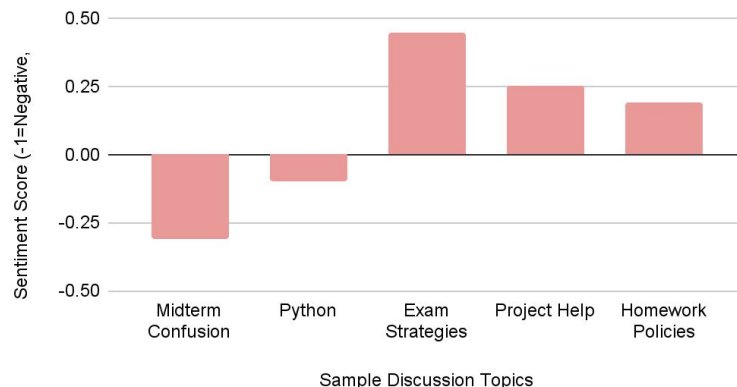
- LDA is a strong baseline for forum topic modeling
 - LDA grouped 3M+ Reddit comments into meaningful clutches like “diet” and “sunscreen,” it was stable over time and interpretable (Cummins & Namburdiri 2022)
- Sentiment analysis adds critical emotional context
 - VADER showed how users felt about each topic in Cummins’ 2024 study
- Synthetic data is valid for model testing
 - Bert models fine-tuned on GPT-Generated data retained performance on classification and clustering tasks (Berenstein 2024)
- Dashboard should show both topic and tone
 - Mello et al. 2024 and Zou et al. 2021 emphasized visualizing engagement with NLP tools

Education Research Recommendations

Goal:

- Incorporate sentiment analysis with topic tracking
- Refine topic modeling with contextual embeddings
- On dashboard, display keyword and sentiment trends over time
- Display interpretable visuals for instructors
- “Human-in-the-loop evaluation”

Simulated VADER Sentiment Scores by Topic



Machine Learning

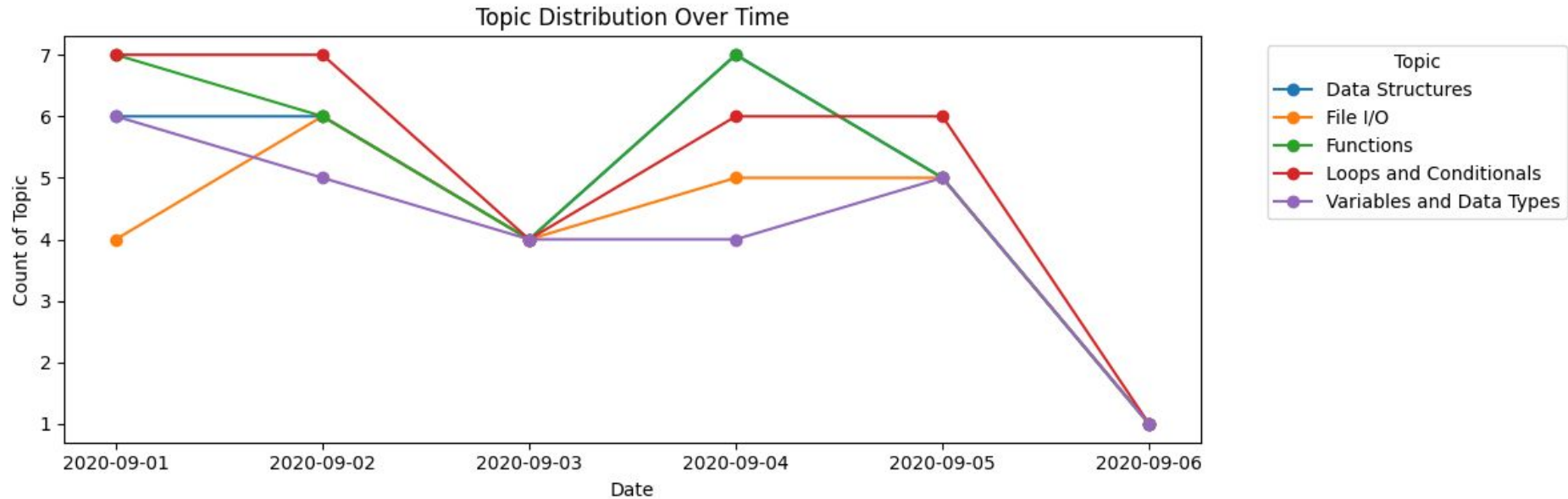
Synthetic Dataset

Synthetic CS 1301 Dataset

- Tested different models & ChatGPT o1's model
- Was able to generate over 140 data points
 - Mimics CS 1301 online discussion form
- Model is randomizing output
 - based on prompt and designated parameters
- Dataset Snippet:

id	Student	Name	Major	Year	ModuleID	Topic	Subtopics	WeeksCovered	LearningOutcomes	StudentFeedback
1	S001	Austin Bates	Computer Science	Freshman	M1	Variables and Data Types	['Integers', 'Strings', 'Booleans']	1	Understand basic data types and variables	I'm unsure when to convert floats to ints for accuracy
2	S002	Isabella Johnson	Mechanical Engineering	Freshman	M2	Loops and Conditions	['For Loops', 'If Statements']	2	Use loops for repetition and conditions	For the midterm, do I need to memorize nested loops?
3	S003	Noah Patel	Electrical Engineering	Sophomore	M3	Functions	['Defining Functions', 'Parameters']	2	Define and use functions in Python	Can we have a function return more than one list with values?
4	S004	Emily Zhang	Computer Science	Freshman	M4	Data Structures	['Lists', 'Dictionaries']	2	Use basic data structures	Is it better to store large sets of data in a dictionary or a list?
5	S005	Jacob Young	Mathematics	Sophomore	M5	File I/O	['Reading Files', 'Writing Files']	1	Read from and write to files in Python	Do we need to handle exceptions for every file operation?
6	S006	Olivia Green	Computer Science	Freshman	M1	Variables and Data Types	['Integers', 'Strings', 'Booleans']	1	Understand basic data types and variables	Is there any real performance difference between int and float?
7	S007	Harper Davis	Mechanical Engineering	Freshman	M2	Loops and Conditions	['For Loops', 'If Statements']	2	Use loops for repetition and conditions	How do I break out of multiple nested loops without using break?
8	S008	Sarah Martin	Mathematics	Sophomore	M3	Functions	['Defining Functions', 'Parameters']	2	Define and use functions in Python	Would using *args for variable-length parameters be better?
9	S009	Daniel Kim	Electrical Engineering	Freshman	M4	Data Structures	['Lists', 'Dictionaries']	2	Use basic data structures	Is it possible that we get tested on dictionary comprehensions?
10	S010	Grace Thompson	Computer Science	Sophomore	M5	File I/O	['Reading Files', 'Writing Files']	1	Read from and write to files in Python	Is 'with open' mandatory, or can I just open and close files manually?
11	S011	Henry White	Computer Science	Junior	M1	Variables and Data Types	['Integers', 'Strings', 'Booleans']	1	Understand basic data types and variables	How do I handle a boolean variable for a flag in an if statement?
12	S012	Ava Roberts	Industrial Engineering	Senior	M2	Loops and Conditions	['For Loops', 'If Statements']	2	Use loops for repetition and conditions	Does indentation have a big effect on partial credit in loops?
13	S013	Mia Garcia	Computer Science	Senior	M3	Functions	['Defining Functions', 'Parameters']	2	Define and use functions in Python	Are lambda functions allowed on the test or should we stick to def?
14	S014	Ryan Foster	Civil Engineering	Sophomore	M4	Data Structures	['Lists', 'Dictionaries']	2	Use basic data structures	I keep forgetting the difference between list.pop() and list.remove()
15	S015	Stella Reyes	Computer Science	Junior	M5	File I/O	['Reading Files', 'Writing Files']	1	Read from and write to files in Python	If we read a file line by line, will that appear on the final exam?

Topic Distribution Over Time (Synthetic)



Data Science

Recap: Goals & Timeline for the Semester

- Explore **text-analysis models** for topic modeling & sentiment analysis
 - **LDA** - Latent Dirichlet Allocation - probabilistic model that identifies common topics in docs
 - **BERT** - Google's NLP model that can learn a wide breadth of tasks relating to language
- Learn **data cleaning & processing** techniques
- Revamp the **dashboard** from last semesters - make it interactive and informative for instructors

Now! 



LDA Model on Synthetic CS 1301 Dataset

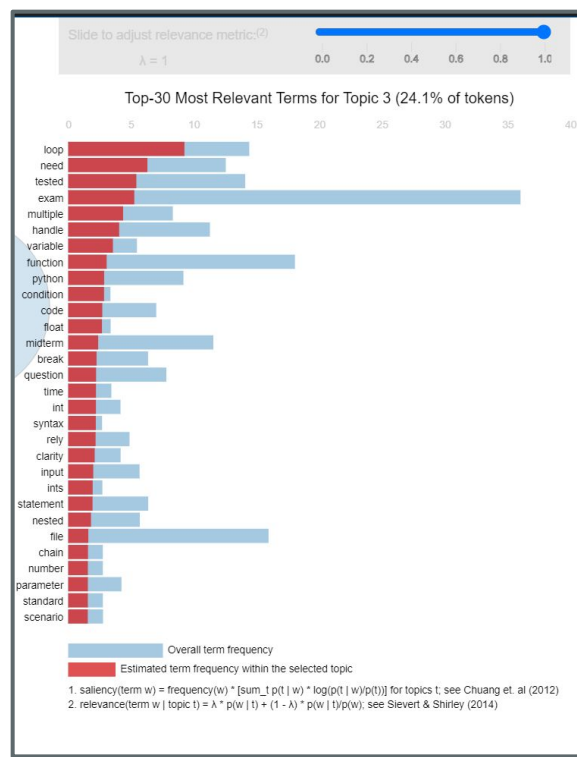
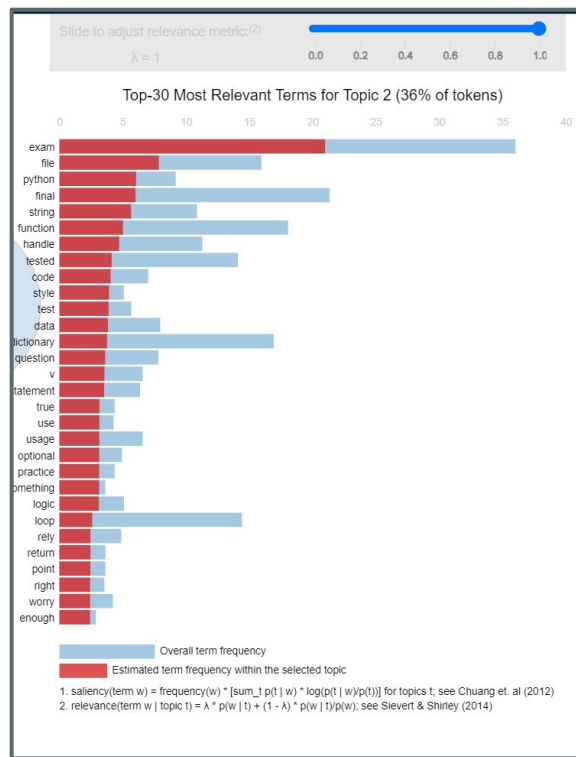
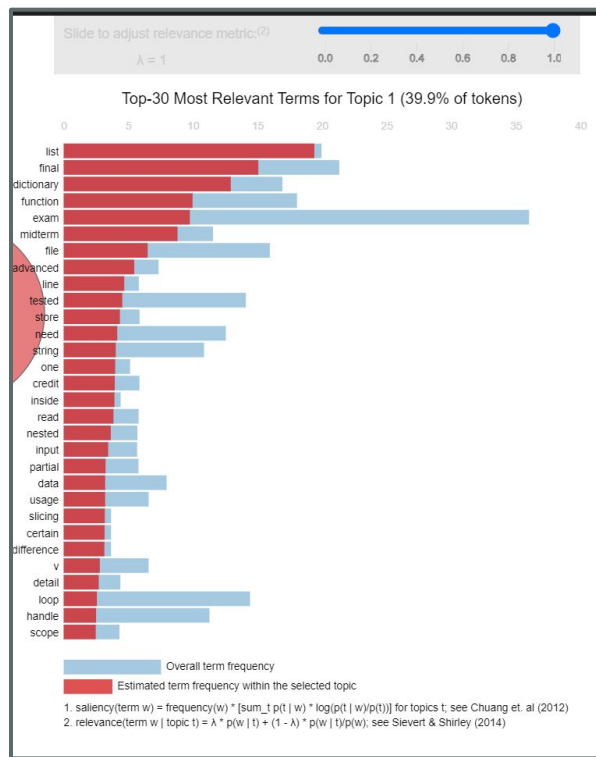
- Using *PyLDAvis Python library*, identified **3 distinct topics** within dataset



- Intertopic Distance Map
 - Shows how different the topics are from each other
 - 3 distinct clusters with minimal overlap
 - Topic 1 is largest ($\approx 40\%$)
 - Topic 2 is next ($\approx 36\%$)
 - Topic 3 is smallest ($\approx 24\%$), but still substantial

LDA Model on Synthetic CS 1301 Dataset

- Top-30 Most Relevant Terms for each topic



LDA Model on Synthetic CS 1301 Dataset

- Topic 1: Assessment-related queries

- Top terms: *list, final, dictionary, function, exam, midterm, store, string, credit, read, input, score*

- Topic 2: Coding Concepts & Syntax Issues

- Top terms: *exam, file, python, string, function, tested, code, style, test, logic, true, return, optional*

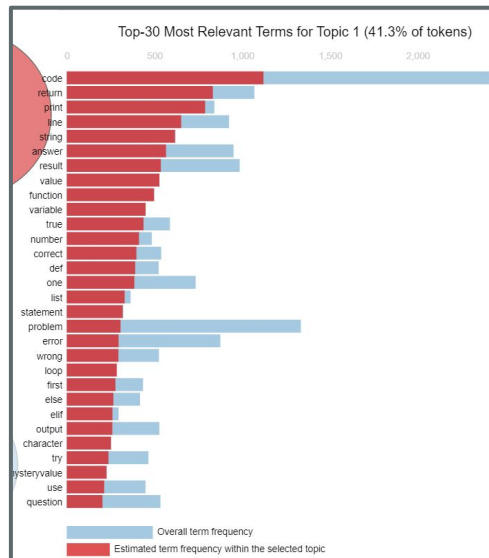
- Topic 3: Debugging Logic & Control Flow & Exam Confusion

- Top terms: *loop, need, texted, exam, variable, condition, syntax, midterm, chain, parameter, scenario*

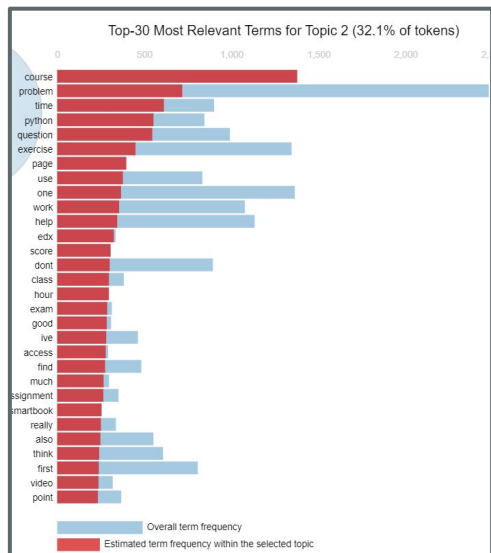
- Analysis

- Term **exam** appears in 3 topics with high frequency
- Topic 2 hints at students asking about how to write or understand Python code
- Topic 3 suggests programming logic confusion possibly tied to exam prep stress

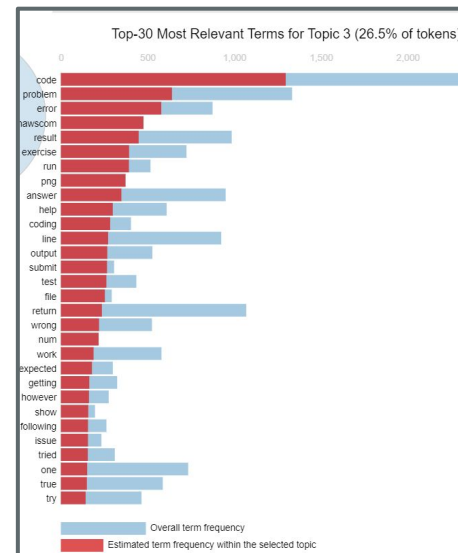
LDA Model on Actual CS 1301 Ed Dataset



Topic 1: Key Programming Concepts



Topic 2: General Course Logistics and Assignments

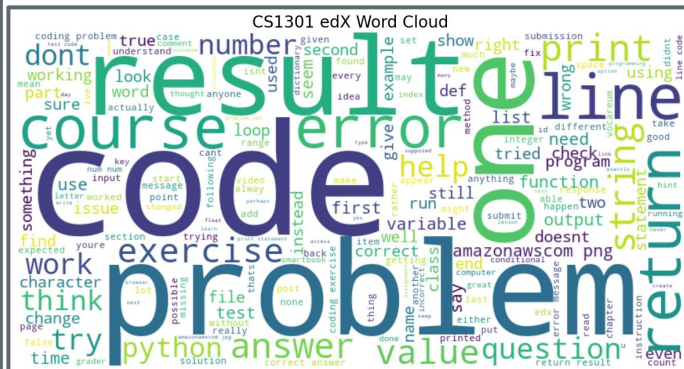
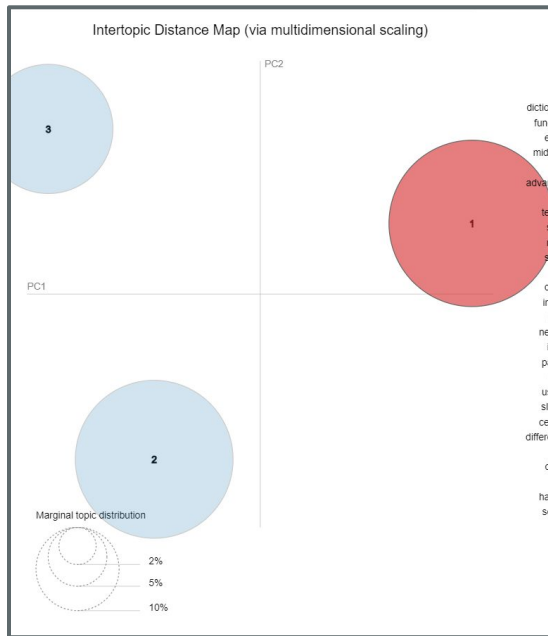


Topic 3: Code Debugging and Troubleshooting

Side-by-Side Comparison

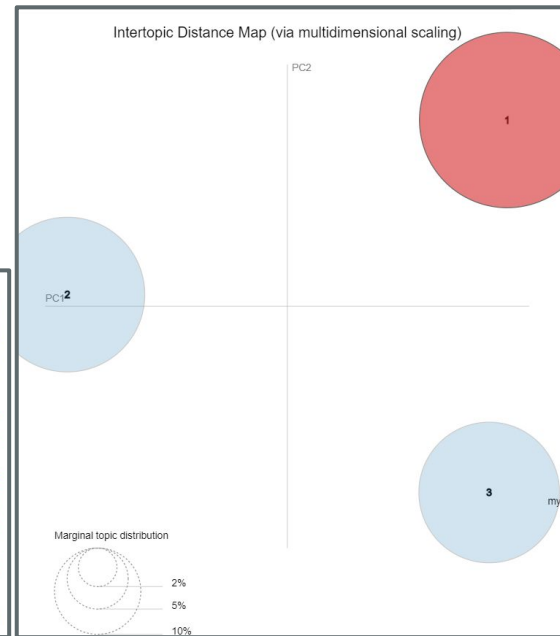
Synthetic

- 141 data entries
- Mockup Queries are more consistent



Actual

- 6467 data entries
- Tons of “non-CS-related” entries



Dashboard

- A React frontend to give user interactive visualization
- Comparing the results from real data and synthetic data

Discussion Forum - Ed Discussion Topic Analysis

Topic Selection

CS 1301 - topic 1

CS 1301 - topic 2

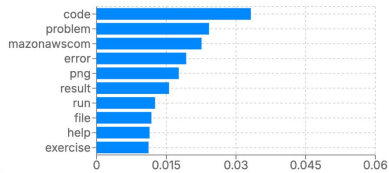
CS 1301 - topic 3

CS 1301 - topic 4

CS 1301 - topic 5

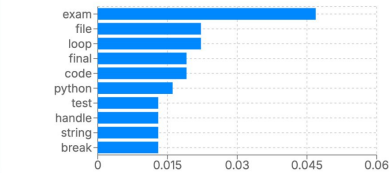
Analysis with Real Dataset

Top Words for CS 1301 - topic 1:



Analysis with Synthetic Dataset

Top Words for Synthetic data - topic 1:



Explanation of Dashboard

This is the dashboard for visualizing the output for discussion forum with LDA topic modeling.

Left side shows the result from real dataset while the right side shows the result from synthetic dataset we generated from our machine learning model.

Discussion Forum - Ed Discussion Topic Analysis

Topic Selection

CS 1301 - topic 1

CS 1301 - topic 2

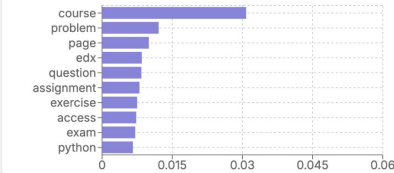
CS 1301 - topic 3

CS 1301 - topic 4

CS 1301 - topic 5

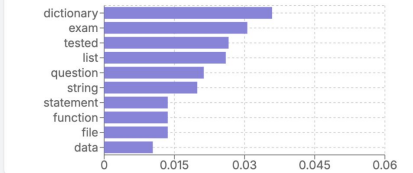
Analysis with Real Dataset

Top Words for CS 1301 - topic 5:



Analysis with Synthetic Dataset

Top Words for Synthetic data - topic 5:



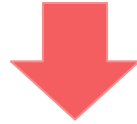
Explanation of Dashboard

This is the dashboard for visualizing the output for discussion forum with LDA topic modeling.

Left side shows the result from real dataset while the right side shows the result from synthetic dataset we generated from our machine learning model.

Conclusion

- Prepared datasets that suitable for analysis
- Generated synthetic dataset
- Performed Topic Modeling



Demonstrates the feasibility and educational value of transforming Ed Discussion forum data into a meaningful tool for educators to improve teaching strategies and course designs.

Next Step...

- Improve model performance
 - Advanced models, e.x. BERTopic
- Incorporate with more courses for the synthetic dataset
 - additional data analysis
- Enhance interactive dashboards

Thanks for listening!

Q&A

Citation

- Berenstein, D. (2024). Fine-tune ModernBERT on synthetic data. Hugging Face Blog.
<https://huggingface.co/blog/davidberenstein1957/fine-tune-modernbert-on-synthetic-data>
- Peng, Jian-E., and Yuanlan Jiang. "Mining Opinions on LMOOCs: Sentiment and Content Analyses of Chinese Students' Comments in Discussion Forums." *System (Linköping)*, vol. 109, 2022, pp. 102879-, <https://doi.org/10.1016/j.system.2022.102879>.
- Jelodar, H., Wang, Y., Orji, R., & Huang, S. (2020). *Deep Sentiment Classification and Topic Discovery on Novel Coronavirus or COVID-19 Online Discussions: NLP Using LSTM Recurrent Neural Network Approach*. *IEEE Journal of Biomedical and Health Informatics*, 24(10), 2733-2742.
- Liu, Sannyuya, et al. "Automated Detection of Emotional and Cognitive Engagement in MOOC Discussions to Predict Learning Achievement." *Computers and Education*, vol. 181, 2022, pp. 104461-, <https://doi.org/10.1016/j.compedu.2022.104461>.
- Cummins, J. A., & Nambudiri, V. E. (2022). *Natural language processing: A window to understanding skincare trends*. *International Journal of Medical Informatics*, 160, 104705. <https://doi.org/10.1016/j.ijmedinf.2022.104705>
- Grootendorst, M. (2023). BERTopic: Topic modeling with BERT embeddings. *BERTopic Documentation & Tutorials*.
<https://maartengr.github.io/BERTopic/>
- Zou, W., Hu, X., Pan, Z., Li, C., Cai, Y., & Liu, M. (2021). Exploring the relationship between social presence and learners' prestige in MOOC discussion forums using automated content analysis and social network analysis. *Computers in Human Behavior*, 115, 106582. <https://doi.org/10.1016/j.chb.2020.106582>